

Lecture 6

Diagonalization, Cayley–Hamilton, Functions of Operators

Throughout, $T \in \mathcal{L}(V)$ and $\chi_T(\lambda) = \det(\lambda I - T)$ is the characteristic polynomial. Tags mark the flavour; a ★ marks a harder one.

Diagonalization

Exercise 6.1 (computational) Diagonalize $A = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$: find P and D with $P^{-1}AP = D$.

Exercise 6.2 (computational) Decide whether $A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ is diagonalizable, with reasons.

Exercise 6.3 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ (eigenvalues 3, 1, eigenvectors $(1, 1)$, $(1, -1)$) compute A^n for general n .

Exercise 6.4 (computational) Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ over $\mathbb{R}$: find P and D with $P^{-1}AP = D$.

Exercise 6.5 (computational) Diagonalize $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ over $\mathbb{C}$.

Exercise 6.6 (computational) Decide whether $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ is diagonalizable, with reasons.

Exercise 6.7 (computational) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ compute A^n for general $n \geq 0$.

Exercise 6.8 (proof) Suppose $T \in \mathcal{L}(V)$ is diagonalizable and has only one eigenvalue λ . Prove that $T = \lambda I$.

Cayley–Hamilton

Exercise 6.9 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Exercise 6.10 (computational) Use the Cayley–Hamilton theorem to express A^{-1} as a polynomial in A for $A = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

Exercise 6.11 (computational) For $A = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, use Cayley–Hamilton to write A^4 in the form $\alpha A + \beta I$, and give the resulting matrix.

Exercise 6.12 (computational) For $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ compute χ_A and use Cayley–Hamilton to express A^{-1} as a polynomial in A ; give A^{-1} explicitly.

Exercise 6.13 (proof) Suppose $T \in \mathcal{L}(V)$ is invertible on a complex space of dimension n . Prove that T^{-1} is a polynomial in T .

Exercise 6.14 (proof) Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. By direct computation show that $A^2 - (\text{tr } A)A + (\det A)I = 0$, thereby verifying Cayley–Hamilton for every 2×2 matrix.

Functions of operators

Exercise 6.15 (computational) For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ compute e^{tA} .

Exercise 6.16 (proof) Suppose T is diagonalizable with $T = PDP^{-1}$. Prove that for any polynomials p, q , the operators $p(T)$ and $q(T)$ commute.

Exercise 6.17 (computational) Let $A = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$. Compute e^{tA} and describe the solution curves of $\dot{x} = Ax$.

Exercise 6.18 (★) Suppose T is diagonalizable with eigenvalues in $\{0, 1\}$ (a projection-like operator). Prove that $T^2 = T$.

Exercise 6.19 (★) Suppose A is diagonalizable and every eigenvalue of A satisfies $|\lambda| < 1$. Prove that $A^n \rightarrow 0$ as $n \rightarrow \infty$ (entrywise).

Exercise 6.20 (computational) Write $A = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ as $A = D + N$ with D diagonalizable, N nilpotent, $DN = ND$, and use it to compute e^{tA} .

Exercise 6.21 (computational) For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ compute e^{tA} .

Exercise 6.22 (computational) Find the positive square root $\sqrt{A}$ of $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ (the one with positive eigenvalues), so that $(\sqrt{A})^2 = A$.

Exercise 6.23 (computational) Compute e^{tA} for $A = \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}$.

Exercise 6.24 (computational) Solve $\dot{x} = Ax$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $x(0) = (2, 0)$.

Exercise 6.25 (computational) Compute e^{tA} for $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Exercise 6.26 (computational) Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with $\text{spec}(A) = \{1, 3\}$. Using spectral mapping, find $\text{spec}(B)$ for $B = A^3 - 2A + I$.

Exercise 6.27 (★) On the spin-1 representation the raising operator acts by $J_+|-1\rangle = \sqrt{2}|0\rangle$, $J_+|0\rangle = \sqrt{2}|1\rangle$, $J_+|1\rangle = 0$. Write its matrix in the basis $(|1\rangle, |0\rangle, |-1\rangle)$ and show J_+ is nilpotent.

Exercise 6.28 (proof) Let A be diagonalizable. Prove that $\det(e^A) = e^{\text{tr} A}$.